Daily Dough: HOA01 to HOA07 Complete Guide & Progress Tracker

Course: ITE 7 – Application Development and Emerging Technologies
Institution: University of Rizal System (URS), Binangonan Campus – College of Computer Studies
Subject Instructor: Anjeline N. Legaspi, LPT, MSIT
Baseline Academic Manual: ITE 7 Module.pdf
Project Title: Daily Dough: An On-Demand E-Commerce Platform for Donut Ordering and Delivery

---

1. Project Overview & Architecture

Daily Dough is an on-demand mobile-commerce (M-Commerce) platform designed for boutique bakery storefronts specializing in artisanal donuts and specialty coffee. The platform bridges customer ordering convenience with streamlined administrative management through two synchronized client applications connected to a single cloud database:

1. **Customer Mobile Application:** Developed with Flutter & Dart, supporting registration, catalog browsing, isolated carts, 3-step checkout, and real-time order tracking.
2. **Admin Web Application:** Developed with HTML5, CSS3, and JavaScript, providing live sales analytics, product catalog CRUD, customer monitoring, and order status lifecycle updates.

> **Architectural Choice: Cloud Firestore vs. Firebase Realtime Database**
> While the ITE 7 Module describes Cloud Firestore, Daily Dough utilizes Firebase Realtime Database (RTDB). Both are official Google Firebase NoSQL products. Realtime Database was selected because its flat JSON tree structure delivers ultra-low latency updates ideal for on-demand food delivery order status tracking.

2. HOA01 – M-Commerce System and Architecture Setup

Module Reference: Pages 10–15 | Status: COMPLETED

Objective: Understand system architecture, install development tools, initialize Flutter, connect Firebase, organize folder structure, and create a Git repository.

Required Output	Specification	Status	Evidence
Running Flutter App	Flutter SDK & default app execution	COMPLETED	`mobile_app/lib/main.dart` (Flutter 3.13.5 / 3.47)
Firebase Connected App	Establish cloud backend connectivity	COMPLETED	`FirebaseService.checkFirebaseConnection()`
Organized Folder Structure	Modular lib/ organization	COMPLETED	`models/`, `screens/`, `services/`

Required Output	Specification	Status	Evidence
GitHub Repository	Version control setup & commit	COMPLETED	Branch `main` with clean `.gitignore`

3. HOA02 – Database Structuring and Firebase Configuration

Module Reference: Pages 16–20 | Status: COMPLETED

Objective: Configure cloud authentication, define database schemas, implement Login/Register screens, and enforce Role-Based Access Control (RBAC).

Required Output	Specification	Status	Evidence
Functional Login/Register	Forms with validation & password toggle	COMPLETED	`login_screen.dart`, `register_screen.dart`
Database Entities	Users, Products, and Orders schema	COMPLETED	RTDB nodes: `/users`, `/customers`, `/products`, `/transactions`
Role-Based Access Control	Customer vs. Administrator separation	COMPLETED	Mobile app blocks admin; Web dashboard blocks customers

4. HOA03 – Mobile Application Development

Module Reference: Pages 21–25 | Status: COMPLETED

Objective: Build the customer storefront: product listing, product details with quantity selector, user-isolated cart, and multi-step checkout.

Required Output	Specification	Status	Evidence
Product Listing	Dynamic grid with search & category filter	COMPLETED	`products_screen.dart` (Donuts & Coffee tabs)
Product Details	Description, price & quantity selector	COMPLETED	`product_detail_screen.dart` (+ / - controls)
Functional Cart System	User-isolated cart & auto-sum	COMPLETED	`cart_screen.dart` scoped to `/carts/{username}`
Checkout Process	Address input, payment selection, order commit	COMPLETED	`checkout_screen.dart` (COD, GCash, Card)

5. HOA04 – Web-Based Admin Dashboard

Module Reference: Pages 26–28 | Status: COMPLETED

Objective: Construct an administrative web dashboard for KPI analytics, product CRUD operations, and order lifecycle management.

Required Output	Specification	Status	Evidence
Functional Admin Dashboard	Responsive UI with live KPI metrics	COMPLETED	`web_dashboard/index.html` running on port 5050
Product CRUD	Create, Read, Update, and Delete products	COMPLETED	`app.js` product form, table, edit modal & delete
Order Monitoring	View orders & update progress status	COMPLETED	Status dropdown: Pending → Processing → Delivered
User Monitoring	View customer accounts & order counts	COMPLETED	`#viewCustomers` customer monitoring table

6. HOA05 – System Integration

Module Reference: Pages 29–32 | Status: `COMPLETED`

Objective: Synchronize mobile and web applications via the cloud database, enforce data validation, and resolve integration issues.

Required Output	Specification	Status	Evidence
Fully Integrated System	Shared cloud database between apps	COMPLETED	Both apps interact with `dailydough-5480d-default-rtdb`
Real-Time Synchronization	Product and order sync across clients	COMPLETED	Verified by `test/admin_added_product_test.dart`
Debugging Report	Formal documentation of bug fixes	COMPLETED	`LAB08_VERIFICATION_REPORT.md` (All 3 bugs resolved)

7. HOA06 – Testing, Optimization, and Deployment

Module Reference: Pages 33–35 | Status: `COMPLETED`

Objective: Execute functional QA testing, collect user survey feedback, compile release Android APK, and deploy admin dashboard.

Required Output	Specification	Status	Evidence
QA Checklist	Module verification matrix (Yes/No)	COMPLETED	`documents/Daily_Dough_QA_Checklist_and_User_Feedback.md`
User Testing Results	Peer review survey & usability score	COMPLETED	Documented in `Daily_Dough_QA_Checklist_and_User_Feedback.md` (4.8/5.0)
Release APK File	Optimized standalone Android package	COMPLETED	`mobile_app/build/app/outputs/flutter-apk/app-release.apk` (55.7 MB)

Required Output	Specification	Status	Evidence
Deployed Dashboard	Web deployment configuration	COMPLETED	`.github/workflows/deploy.yml` (GitHub Pages) + local server

8. HOA07 – Final System Presentation and Defense

Module Reference: Pages 36–38 | Status: ***COMPLETED***

Objective: Create presentation slides, master live demonstration sequence, and prepare oral defense reflection answers.

Required Output	Specification	Status	Evidence
Presentation Slides	9-section slide deck with speaker notes	COMPLETED	`documents/Daily_Dough_Final_Presentation_and_Defense_Guide.md`
Demonstration Script	8-step live walkthrough sequence	COMPLETED	Documented in `Daily_Dough_Final_Presentation_and_Defense_Guide.md`
Reflection Answers	Answers for 3 defense questions	COMPLETED	Documented in `Daily_Dough_Final_Presentation_and_Defense_Guide.md`

9. Final Project Submission Master Checklist

Module Reference: Page 38 | Status: 100% COMPLETE

- ✅ **1. Mobile Application:** `daily-dough/mobile_app/`
- ✅ **2. Web Admin Dashboard:** `daily-dough/web_dashboard/`
- ✅ **3. Firebase Integration:** `https://dailydough-5480d-default-rtdb.firebaseio.com/`
- ✅ **4. GitHub Repository:** Local Git repository on branch `main` with deployment workflow
- ✅ **5. Documentation:** `documents/Daily_Dough_Project_Documentation.md`
- ✅ **6. Presentation Slides:** `documents/Daily_Dough_Final_Presentation_and_Defense_Guide.md`
- ✅ **7. APK File:** `mobile_app/build/app/outputs/flutter-apk/app-release.apk` (55.7 MB)